SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE: DSA0204**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 10%

Code of Conduct:

I ___Chalamala Sai Harshitha (192424346)___ *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. **Motion Estimation Variance**

   Errors:

Frame	**A**	**B**
1	**2**	**6**
2	**3**	**7**
3	**2**	**5**

   (a) Compute mean error.

   (b) Compare variability and evaluate stability.

(a) Mean Error

The formula for mean error is:

Mean Error = Sum of errors / Number of frames

Method A:

Mean Error = (2 + 3 + 2) / 3

= 7 / 3

= **2.33 pixels**

Method B:

Mean Error = (6 + 7 + 5) / 3

= 18 / 3

= **6 pixels**

Therefore, Method A has a much lower mean error than Method B.

(b) Variability and Stability

Factor	**Method A**	**Method B**
Mean error	2.33 pixels	6 pixels
Minimum error	2	5
Maximum error	3	7
Error range	1 pixel	2 pixels
Stability	Higher	Lower

Method A:

- Errors are 2, 3, and 2 pixels.
- The error range is only 1 pixel.
- The values remain close to the mean of 2.33 pixels.
- This indicates low variability and stable performance.

Method B:

- Errors are 6, 7, and 5 pixels.
- The error range is 2 pixels.
- Its errors are considerably higher than Method A.
- This indicates greater error and comparatively lower stability.

Evaluation

Method A performs better in both accuracy and consistency. Its mean error is only 2.33 pixels, compared with 6 pixels for Method B. It also has a smaller error range.

Although both methods show some variation across frames, Method A maintains a much more consistent error level. Therefore, it is more reliable for applications where accurate and stable motion estimation is required.

Conclusion

Method A is the better method. It has a lower mean error (2.33 pixels) and smaller variability (1 pixel range) compared with Method B, which has a mean error of 6 pixels and a 2-pixel range. Hence, Method A provides more accurate and stable motion estimation and would be preferable for reliable motion-tracking applications.

2. **IOU Comparison**

   IOU values:

Object	**A**	**B**
1	**0.6**	**0.8**
2	**0.7**	**0.85**
3	**0.65**	**0.9**

   (a) Compute average IOU.

   (b) Compare and evaluate detection quality.

(a) Average IoU

The formula is:

Average IoU = Sum of IoU values / Number of objects

Model A:

Average IoU = (0.60 + 0.70 + 0.65) / 3

= 1.95 / 3

= **0.65**

Model B:

Average IoU = (0.80 + 0.85 + 0.90) / 3

= 2.55 / 3

= **0.85**

(b) Comparison

Factor	Model A	Model B
Object 1	0.60	0.80
Object 2	0.70	0.85
Object 3	0.65	0.90
Average IoU	0.65	0.85
Overall detection quality	Lower	Higher

Analysis

Model A has an average IoU of **0.65**, meaning its predicted bounding boxes have a moderate overlap with the actual object regions.

Model B achieves an average IoU of **0.85**, which is considerably higher. It also performs better for **every individual object**:

- Object 1: 0.80 compared with 0.60
- Object 2: 0.85 compared with 0.70
- Object 3: 0.90 compared with 0.65

The difference in average IoU is:

0.85 − 0.65 = 0.20

Thus, Model B provides a 0.20 higher average IoU, indicating substantially better localization.

Evaluation

IoU measures how closely the predicted bounding box overlaps the actual bounding box. A higher IoU indicates better object localization.

Since Model B has a higher IoU for all three objects and an average IoU of 0.85, it provides more accurate and consistent localization than Model A.

Conclusion

Model B is the better detection model. Its average IoU of 0.85 is significantly higher than Model A's 0.65, demonstrating better overlap between predicted and actual object regions. Therefore, Model B provides superior detection and localization quality and would be the preferred model when accurate object localization is required.

3. **Recall Comparison in Detection**

Results:

Model	**TP**	**FN**
A	**60**	**40**
B	**80**	**20**

(a) Compute recall.

(b) Compare and evaluate sensitivity.

(a) Recall Calculation

The formula for recall is:

Recall = TP / (TP + FN)

Model A

Recall = 60 / (60 + 40)

= 60 / 100

= **0.60 = 60%**

Model B

Recall = 80 / (80 + 20)

= 80 / 100

= **0.80 = 80%**

(b) Comparison

Factor	**Model A**	**Model B**
True Positives	60	80
False Negatives	40	20
Recall	60%	80%
Sensitivity	Lower	Higher
Missed detections	More	Fewer

Analysis

Model A has a recall of 60%, meaning it correctly detects 60% of the actual positive cases. However, it misses 40% of the positive cases, resulting in a higher number of false negatives.

Model B achieves 80% recall and misses only **20%** of the actual positive cases. Therefore, it is more effective at identifying positive instances.

The difference in recall is:

80% − 60% = 20 percentage points

This indicates that Model B has substantially better sensitivity than Model A.

Evaluation

Recall is particularly important when missing a positive object or event is costly. A higher recall means that the detection system can identify a larger proportion of actual positive cases.

Since Model B has both higher true positives and fewer false negatives, it provides better sensitivity and is more reliable for applications where missed detections should be minimized.

Conclusion

Model B performs better than Model A. It achieves a recall of 80%, compared with 60% for Model A. Its higher recall and lower number of false negatives indicate better sensitivity. Therefore, Model B is the preferred model when detecting as many actual positive cases as possible is the main objective.

4. **Depth Estimation Error (Squared)**

   Predicted:

Object	**A**	**B**	**Actual**
1	**2.5**	**2.2**	**2**
2	**3.5**	**3.1**	**3**
3	**4.5**	**4.2**	**4**

   (a) Compute squared error.

   (b) Compare and evaluate accuracy.

(a) Squared Error Calculation

The formula is:

Squared Error = (Predicted Value − Actual Value)²

Object 1

Method A:

$(2.5 - 2.0)^2 = (0.5)^2 = \mathbf{0.25\ m^2}$

Method B:

$(2.2 - 2.0)^2 = (0.2)^2 = \mathbf{0.04\ m^2}$

Object 2

Method A:

$(3.5 - 3.0)^2 = (0.5)^2 = \mathbf{0.25\ m^2}$

Method B:

$(3.1 - 3.0)^2 = (0.1)^2 =$ **0.01 m²**

Object 3

Method A:

$(4.5 - 4.0)^2 = (0.5)^2 =$ **0.25 m²**

Method B:

$(4.2 - 4.0)^2 = (0.2)^2 =$ **0.04 m²**

Squared Error Comparison

Object	Method A	Method B
1	0.25 m²	0.04 m²
2	0.25 m²	0.01 m²
3	0.25 m²	0.04 m²
Total Squared Error	**0.75 m²**	**0.09 m²**

(b) Evaluation

Method A produces a total squared error of 0.75 m², while Method B produces only 0.09 m².

The difference is:

$0.75 - 0.09 = 0.66$ m²

Method B therefore has a substantially lower error. Its predicted depth values are much closer to the actual values for all three objects.

Since lower squared error indicates better prediction accuracy, Method B provides more accurate depth estimation.

Conclusion

Method B is the better method because its total squared error is only 0.09 m², compared with 0.75 m² for Method A. The lower error shows that Method B's depth predictions are consistently closer to the actual values. Therefore, Method B should be preferred when accurate depth estimation is required.

5. **Recognition Precision Comparison**

Results:

System	TP	FP
A	**90**	**10**
B	**85**	**5**

(a) Compute precision.

(b) Compare and evaluate performance.

(a) Precision Calculation

The formula for precision is:

Precision = TP / (TP + FP)

System A

Precision = 90 / (90 + 10)

= 90 / 100

= **0.90 = 90%**

System B

Precision = 85 / (85 + 5)

= 85 / 90

= **0.9444 = 94.44%**

(b) Comparison

Factor	System A	System B
True Positives	90	85
False Positives	10	5
Precision	**90%**	**94.44%**
False detections	Higher	Lower
Precision performance	Good	Better

Analysis

System A correctly identifies 90 positive cases, but it also produces 10 false positives. Its precision is therefore 90%.

System B identifies 85 true positives and produces only 5 false positives. Although it has fewer true positives than System A, its precision is higher at 94.44%.

The difference in precision is:

94.44% − 90% = 4.44 percentage points

This shows that System B is more accurate when considering the proportion of its positive predictions that are actually correct.

Evaluation

System B is preferable when the application needs to minimize false positive detections. Its lower number of false positives results in higher precision.

System A has a higher number of true positives, but this comes with more false positives. Therefore, System A may be useful when detecting more positive cases is important, while System B is better when prediction reliability and fewer false alarms are the priority.

Conclusion

System B performs better in terms of precision. It achieves 94.44% precision, compared with 90% for System A, because it produces fewer false positives. Therefore, System B is the preferred system when reliable positive predictions and reduced false detections are important.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand